



NORTH SYDNEY BOYS HIGH SCHOOL

2026 HSC ASSESSMENT TASK 3 (TRIAL HSC)

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write in the space provided in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Extra working space is provided at the end of examination.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Lin
- ☐ Ms Everingham
- ☐ Ms Cai
- ☐ Ms Sarofim

(To be used by the exam markers only.)

Question Numbers	1-10	11-16	17-21	22-25	26-29	30-32	33-35	Total
Mark	$\overline{10}$	$\overline{14}$	$\overline{16}$	$\overline{14}$	$\overline{15}$	$\overline{16}$	$\overline{15}$	$\overline{100}$

Section I

10 Marks

Attempt Questions 1-10

Allow about 15 minutes for this section

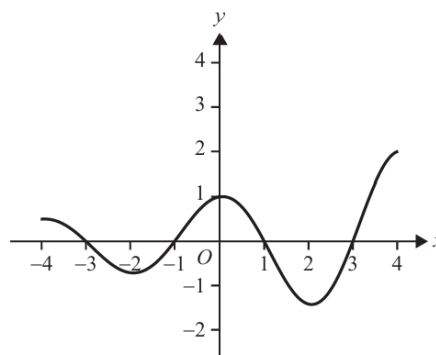
Use the multiple-choice sheet for Questions 1-10

- 1 What is the 20th term of the following arithmetic sequence

$$-7, -3, 1, 5, \dots$$

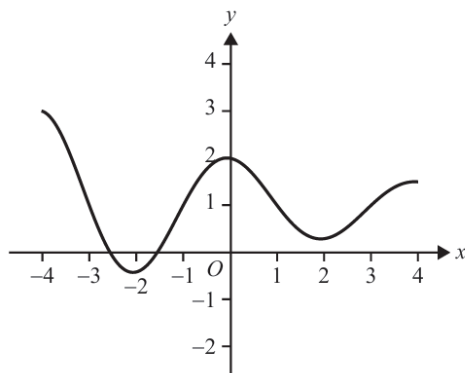
- A. 65
- B. 69
- C. 73
- D. 77

- 2 The graph of $y = f(x)$ is shown below.

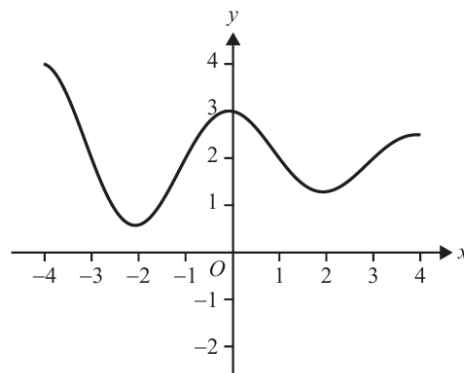


Which of the following options best represents the graph of $y = f(-x) + 2$?

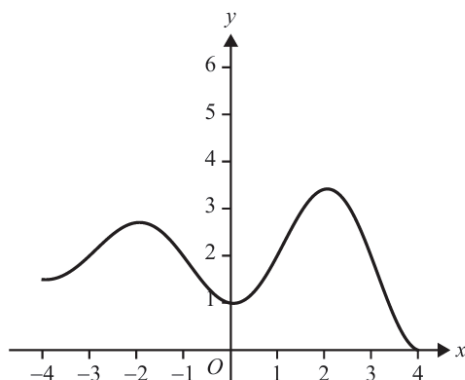
A.



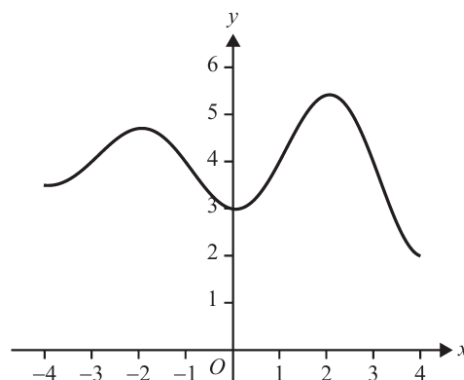
B.



C.



D.



- 3 The equation $x^2 + y^2 - 2x + 6y + 6 = 0$ represents a circle with centre C and radius r

Which of the following is correct?

- A. $C = (1, -3)$ and $r = 2$
- B. $C = (1, -3)$ and $r = 4$
- C. $C = (-1, 3)$ and $r = 2$
- D. $C = (-1, 3)$ and $r = 4$

- 4 Which of the following is equivalent to $-\log_a \left(\frac{x}{y}\right) - \log_a y$?

- A. $\log_a \left(\frac{1}{x}\right)$
- B. $\log_a \left(\frac{1}{y}\right)$
- C. $\log_a \left(\frac{x}{y^2}\right)$
- D. $\log_a \left(\frac{y}{x}\right)$

- 5 A fair spinner numbered 1 to 12 is spun once.

Let A be the event that the number spun is odd, and let B be the event that the number spun is greater than 6.

Which statement is true?

- A. A and B are independent.
- B. A and B are mutually exclusive.
- C. A and B are neither independent nor mutually exclusive.
- D. A and B are both independent and mutually exclusive.

- 6 A camera uses the exposure value $E = \log_2 \left(\frac{1}{t}\right)$ where t is the shutter time.

If the shutter time is halved, E increases by

- A. 0.5
- B. 1
- C. 2
- D. $\log_2(2t)$

7 What is the number of solutions to the equation $\sin 2x (\sec x - 1) = 0$ for $0 \leq x \leq 2\pi$

A. 2

B. 3

C. 5

D. 7

8 One day, at a particular school, m students walked to school and the remaining n students travelled to school using a different form of transport.

Of the m students who walked, 20% took at least 30 minutes to get to school.

Of the n students who used a different form of transport, 40% took at least 30 minutes to get to school.

Given that a randomly selected student took at least 30 minutes to get to school, the probability they walked to school is given by

A. $\frac{m}{m+n}$

B. $\frac{2n}{m+n}$

C. $\frac{m}{5(m+n)}$

D. $\frac{m}{m+2n}$

9 Let $f(x) = ae^{kx}$, where $a > 0$ and $k > 0$

Suppose that $f(p) = 3$ and $f(q) = 12$.

What is the value of $f\left(\frac{p+q}{2}\right)$?

A. 6

B. 7.5

C. 6.5

D. Cannot be determined with the given information.

10 What is the value of $\int_0^3 f(x) dx$ given that $f(x) + f(3-x) = 10$, for all $x \in [0, 3]$?

A. 10

B. 15

C. 20

D. 30

SECTION II

90 Marks
Attempt Questions 11-35
Allow about 2 hours and 45 minutes for this section

Question 11

Find the sum of the following arithmetic series **2**

$$2 + 5 + 8 + \dots + 53$$

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Question 12

Solve $|2x - 5| = 3$ **2**

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Question 13**2**

Solve $2x^2 + 5x - 12 > 0$

Question 14

Find the equation of the tangent to the curve $f(x) = 6\sqrt{x+1} + 5$ at the point where $x = 8$

3

Question 15

Stations A and B are 40km apart where station B is due east of A .

A flare at point F is observed on a bearing of $055^\circ T$ from A and a bearing of $315^\circ T$ from B .

- (a)** Sketch a diagram showing the above information.

1

- (b)** Find the distance from Station A to the flare.

2

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 16**2**

Differentiate $f(x) = x^2 \cos x$

Question 17**3**

Given $f(x) = \sqrt{x+1}$ and $g(x) = 2x - 3$.

Find the domain and range of $f(g(x))$.

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This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 19

Adrian picks two notes at random from five \$5, three \$10 and two \$20 notes.

Find the probability that Adrian has picked a total of at least \$20.

3

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 20

Evaluate $\int_0^1 \frac{4x+2}{x^2+x+1} dx$.

2

[illegible]

Question 21

(a) Sketch the curve $y = \ln(x + 1)$, showing all important features.

2

(b) Using the trapezoidal rule with 4 subintervals, estimate the value of $\int_1^3 \ln(x+1) dx$ correct to 3 decimal places

2

This image shows a single sheet of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 22

The amount of money gained by an entrepreneur doubles every day. On day 1, she earns \$9 and on day 2 she earns \$18.

3

On which day will her cumulative earnings first exceed \$1,000,000?

[illegible]

Question 23

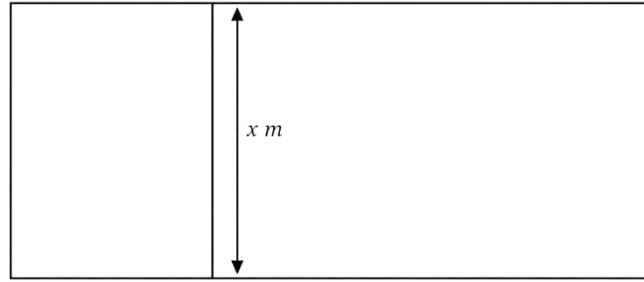
Prove that $\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} = 2 \operatorname{cosec} x$

3

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 24

A farmer has a large patch of land, on which he needs to build two fenced-in, rectangular areas as shown in the diagram.



- (a) Given the farmer has 260m of fencing, show that the total area of the two rectangles is given by

1

$$A = 130x - \frac{3x^2}{2}$$

[illegible]

- (b) Hence or otherwise, find the maximum area of the paddock, showing that it is a maximum area.

3

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 25

In a survey of 200 students.

- 120 play soccer
- 90 play basketball
- 170 play at least one of the two sports.

- (a) By finding the number of students that play both sports. Draw a Venn diagram showing all the relevant information.

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[illegible]

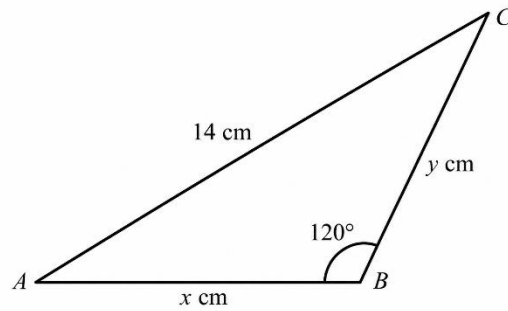
- (b) Two students are randomly selected without replacement. Find the probability that at least one of them plays exactly one of the two sports.

2

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 26

In the triangle ABC below, $AB = x$ cm, $BC = y$ cm. $AC = 14$ cm and angle $ABC = 120^\circ$



- (a)** Show that $x^2 + y^2 + xy = 196$ **1**

[illegible]

- (b)** Given that the perimeter of triangle ABC is 30 cm, find the exact area of triangle ABC **3**

[illegible]

Question 27

Let $f(x) = \frac{1}{e^x + 1}$ and $g(x) = f(x) - \frac{1}{2}$

3

Prove that $g(x)$ is an **odd** function

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 28

Solve $\ln(6e^x + 1) + x = 0$, writing your answer in logarithmic form

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Question 29

A particle, P , travels in a straight line from O such that its velocity, v m/s, at time t seconds after passing O is given by $v = 24 \left(e^{-\frac{t}{6}} - \frac{1}{e} \right)$

- (a) Find the value of t at which the particle P is instantaneously at rest.

2

[illegible]

- (b) Find the total distance travelled by the particle P for the first 9 seconds to the nearest metre.

3

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 30

(a) Show that $\frac{d}{dx} \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right) = \frac{2}{(\sin x + \cos x)^2}$

2

(b) Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin x - \cos x + 2}{(\sin x + \cos x)^2} dx$

3

Question 31

Given the function $f(x) = \ln(x^2 + 1) - x$

- (a) Find the stationary point(s) and determine their nature.

3

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b) Find the point(s) of inflection.

2

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.

(c) Sketch the graph of $y = f(x)$, for $-2 \leq x \leq 2$ showing all important features.

2

1

$$T = 20 - 8 \cos\left(\frac{\pi}{12}(t - 4)\right)$$

Determine the total amount of time each day that the shade cloth is deployed.

[illegible]

Question 33

(a) Sketch the graphs of $y = -2 \cos 3x + 1$ and $y = -\frac{2}{\pi}x + 1$ on the same axes, showing all important features, for $0 \leq x \leq \pi$.

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(b) Explain how the number of solutions for $2 \cos 3x = \frac{2}{\pi}x$ can be found using the graph drawn in part (a)

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Question 34

Let g be the function obtained from f by a sequence of transformations such that

$$g(x) = af(b - x) + c.$$

The point $(2, 5)$ on the graph of $y = f(x)$ corresponds to the point $(7, -11)$ on the graph of $y = g(x)$.

The point $(4, 1)$ on the graph of $y = f(x)$ corresponds to the point $(5, -3)$ on the graph of $y = g(x)$.

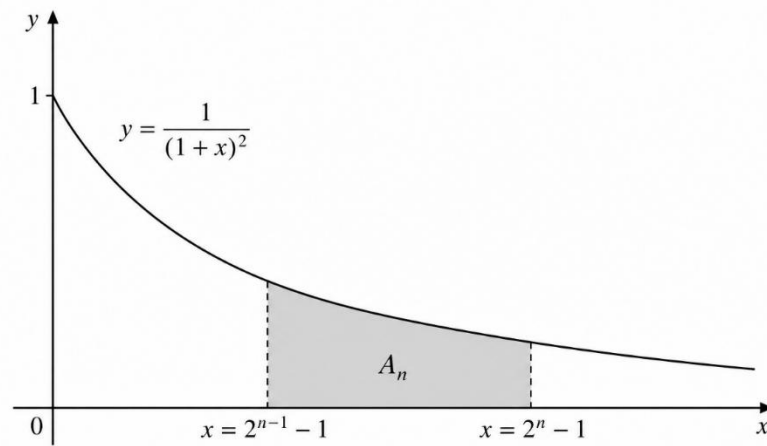
Find the values of a, b and c.

This image shows a full page of a worksheet designed for handwriting practice. It features 20 evenly spaced, horizontal dashed lines across the entire width of the page. The background is plain white, providing a clear guide for letter height and placement. There are no margins, text, or other markings present.

Please turn the page for Question 35.

Let $f(x) = \frac{1}{(1+x)^2}, x \geq 0$.

$x = 2^{n-1} - 1$ and $x = 2^n - 1$ as shown below.



(a) Show that $A_n = \frac{1}{2^n}$

2

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

End of Examination

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

A series of horizontal dotted lines spanning the width of the page, intended as a guide for handwriting practice.

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Mathematics Advanced

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Class Teacher:

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Section I

10 Marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice sheet for Questions 1-10

- 1 What is the 20th term of the following arithmetic sequence

$$-7, -3, 1, 5, \dots$$

- A. 65
B. 69
C. 73
D. 77

$$a = -7$$

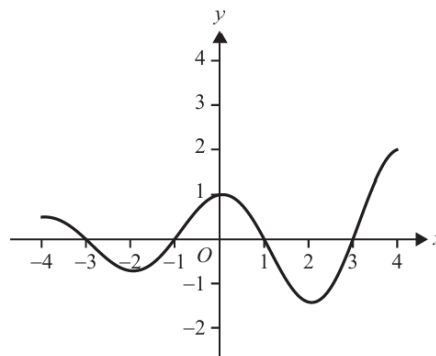
$$d = 4$$

$$T_{20} = a + (n-1)d$$

$$= -7 + 19 \times 4$$

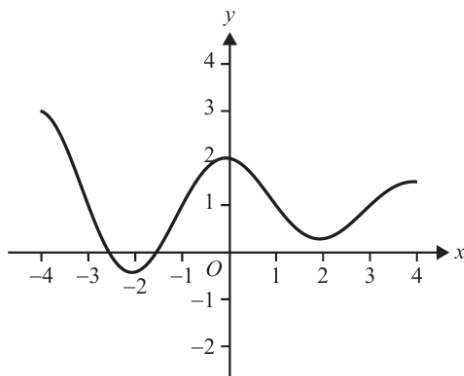
$$= 69$$

- 2 The graph of $y = f(x)$ is shown below.

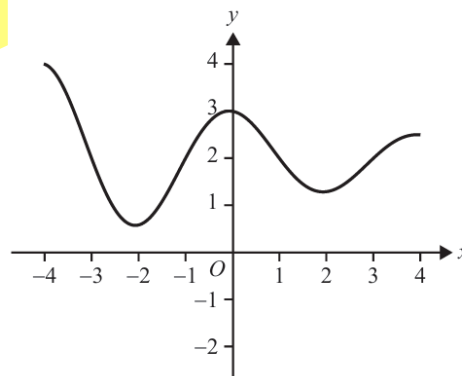


Which of the following options best represents the graph of $y = f(-x) + 2$?

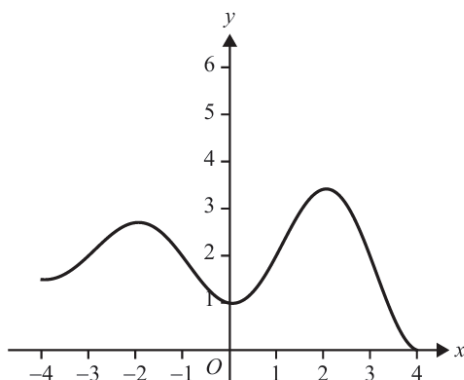
A.



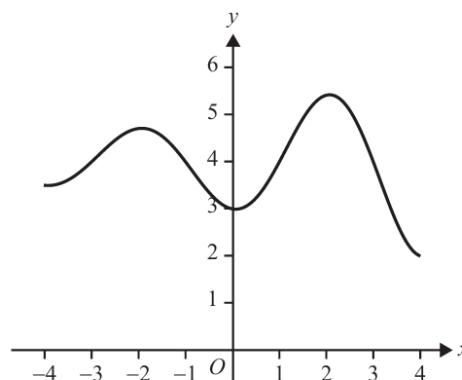
B.



C.



D.



- 3 The equation $x^2 + y^2 - 2x + 6y + 6 = 0$ represents a circle with centre C and radius r

Which of the following is correct?

$$(x-1)^2 + (y+3)^2 = -6 + 1 + 9 \\ = 4$$

- A. $C = (1, -3)$ and $r = 2$
 B. $C = (1, -3)$ and $r = 4$
 C. $C = (-1, 3)$ and $r = 2$
 D. $C = (-1, 3)$ and $r = 4$

- 4 Which of the following is equivalent to $-\log_a \left(\frac{x}{y}\right) - \log_a y$?

A. $\log_a \left(\frac{1}{x}\right)$

$$\log_a \left(\frac{y}{x}\right) - \log_a y$$

B. $\log_a \left(\frac{1}{y}\right)$

$$= \log_a \left(\frac{1}{x}\right)$$

C. $\log_a \left(\frac{x}{y^2}\right)$

D. $\log_a \left(\frac{y}{x}\right)$

- 5 A fair spinner numbered 1 to 12 is spun once.

Let A be the event that the number spun is odd, and let B be the event that the number spun is greater than 6.

Which statement is true?

$$A = \{1, 3, 5, 7, 9, 11\}$$

$$P(A) = \frac{6}{12} = \frac{1}{2}$$

$$B = \{7, 8, 9, 10, 11, 12\}$$

$$P(B) = \frac{6}{12} = \frac{1}{2}$$

- A. A and B are independent.
 B. A and B are mutually exclusive.
 C. A and B are neither independent nor mutually exclusive.
 D. A and B are both independent and mutually exclusive.

$$A \cap B = \{7, 9, 11\}$$

$$P(A \cap B) = \frac{3}{12} = \frac{1}{4}$$

$$P(A) \times P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

- 6 A camera uses the exposure value $E = \log_2 \left(\frac{1}{t}\right)$ where t is the shutter time.

If the shutter time is halved, E increases by

- A. 0.5
 B. 1
 C. 2
 D. $\log_2(2t)$

7 What is the number of solutions to the equation $\sin 2x (\sec x - 1) = 0$ for $0 \leq x \leq 2\pi$

A. 2

B. 3

C. 5

D. 7

$$\begin{aligned} \sin 2x &= 0 & \sec x &= 1 \\ 2x &= 0, \pi, 2\pi, 3\pi, 4\pi & \frac{1}{\cos x} &= 1 \\ x &= 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi & \therefore x &= 0, 2\pi \\ & & \text{but } x &= \frac{\pi}{2}, \frac{3\pi}{2} \\ \therefore x &= 0, \pi, 2\pi \end{aligned}$$

8 One day, at a particular school, m students walked to school and the remaining n students travelled to school using a different form of transport.

Of the m students who walked, 20% took at least 30 minutes to get to school.

Of the n students who used a different form of transport, 40% took at least 30 minutes to get to school.

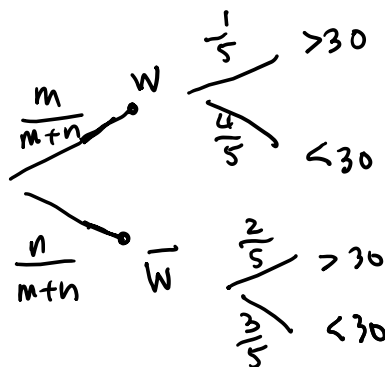
Given that a randomly selected student took at least 30 minutes to get to school, the probability they walked to school is given by

A. $\frac{m}{m+n}$

B. $\frac{2n}{m+n}$

C. $\frac{m}{5(m+n)}$

D. $\frac{m}{m+2n}$



$$\begin{aligned} P(W | >30) &= \frac{P(W \cap >30)}{P(>30)} \\ &= \frac{\frac{m}{m+n} \times \frac{1}{5}}{\frac{m}{m+n} \times \frac{1}{5} + \frac{n}{m+n} \times \frac{2}{5}} \\ &= \frac{m}{m+2n} \end{aligned}$$

9 Let $f(x) = ae^{kx}$, where $a > 0$ and $k > 0$

Suppose that $f(p) = 3$ and $f(q) = 12$.

What is the value of $f\left(\frac{p+q}{2}\right)$?

A. 6

B. 7.5

C. 6.5

D. Cannot be determined with the given information.

$$\begin{aligned} f(p) &= 3 \Rightarrow ae^{kp} = 3 \therefore e^{kp} = \frac{3}{a} \\ f(q) &= 12 \Rightarrow ae^{kq} = 12 \therefore e^{kq} = \frac{12}{a} \\ \text{Now } f\left(\frac{p+q}{2}\right) &= ae^{k\left(\frac{p+q}{2}\right)} \\ &= ae^{\frac{kp}{2}} \cdot e^{\frac{kq}{2}} \\ &= a(e^{kp})^{\frac{1}{2}} \cdot (e^{kq})^{\frac{1}{2}} \\ &= a \cdot \sqrt{\frac{3}{a}} \cdot \sqrt{\frac{12}{a}} \\ &= \sqrt{36} \\ &= 6 \end{aligned}$$

10 What is the value of $\int_0^3 f(x) dx$ given that $f(x) + f(3-x) = 10$, for all $x \in [0, 3]$?

A. 10

B. 15

C. 20

D. 30

$$\begin{aligned} f(3-x) &\Rightarrow f(x) \text{ translated to the left by } 3 \text{ units and flipped around } y \text{ axis.} \\ \text{This is equivalent to flipping around the line } x &= \frac{3}{2} \\ \therefore \int_0^3 f(3-x) dx &= \int_0^3 f(x) dx \\ \therefore \int_0^3 f(x) + f(3-x) dx &= 2 \int_0^3 f(x) dx \\ \therefore \int_0^3 10 dx &= 2 \int_0^3 f(x) dx \\ \therefore \int_0^3 f(x) dx &= 15 \end{aligned}$$

SECTION II

90 Marks

Attempt Questions 11-34

Allow about 2 hours and 45 minutes for this section

Question 11

Find the sum of the following arithmetic series

2

$$2 + 5 + 8 + \dots + 53$$

$$a = 2, d = 3$$

$$\therefore 53 = 2 + (n-1)3$$

$$51 = 3(n-1)$$

$$n-1 = 17$$

$$n = 18$$

$$\therefore S_{18} = \frac{18}{2} (2 + 53)$$

$$= 9 \times 55$$

$$= 495$$

Question 12

Solve $|2x - 5| = 3$

2

$$2x - 5 = \pm 3$$

$$2x = 5 \pm 3$$

$$\therefore 2x = 8 \quad \text{or} \quad 2x = 2$$

$$x = 4 \quad \text{or} \quad x = 1$$

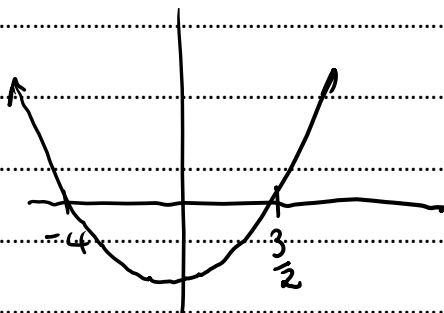
Question 13

2

Solve $2x^2 + 5x - 12 > 0$

$$2x^2 + 5x - 12 > 0$$

$$(2x - 3)(x + 4) > 0$$



$$\therefore x < -4 \quad \text{or} \quad x > \frac{3}{2}$$

Question 14

3

Find the equation of the tangent to the curve $f(x) = 6\sqrt{x+1} + 5$ at the point where $x = 8$

$$f'(x) = \frac{3}{\sqrt{x+1}}$$

$$f'(8) = \frac{3}{3} = 1$$

$$f(8) = 6\sqrt{9} + 5 \\ = 23$$

$\therefore$ Equation of tangent:

$$y - 23 = 1(x - 8)$$

$$y = x - 8 + 23$$

$$= x + 15$$

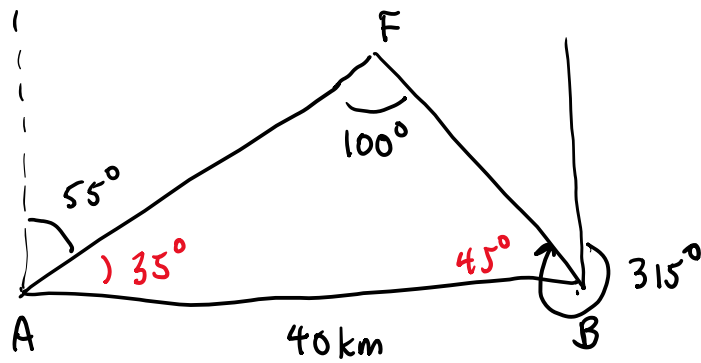
Question 15

Stations A and B are 40km apart where station B is due east of A .

A flare at point F is observed on a bearing of $055^\circ T$ from A and a bearing of $315^\circ T$ from B .

(a) Sketch a diagram showing the above information.

1



(b) Find the distance from Station A to the flare.

2

$$\frac{AF}{\sin 45^\circ} = \frac{40}{\sin 100^\circ}$$

$$AF = \frac{40 \sin 45^\circ}{\sin 100^\circ}$$

$$\approx 28.7 \text{ km}$$

Question 16

2

Differentiate $f(x) = x^2 \cos x$

$$u = x^2 \quad v = \cos x$$

$$u' = 2x \quad v' = -\sin x$$

$$\therefore f'(x) = 2x \cos x - x^2 \sin x$$

Question 17

3

Given $f(x) = \sqrt{x+1}$ and $g(x) = 2x - 3$.Find the domain and range of $f(g(x))$.

$$f(g(x)) = \sqrt{(2x-3)+1}$$

$$= \sqrt{2x-2}$$

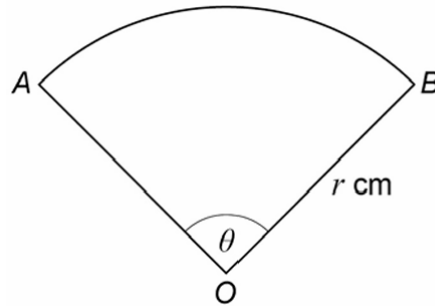
$$\therefore 2x - 2 \geq 0$$

$$\therefore D: x \geq 1 \quad R: y \geq 0$$

Question 18

The diagram below shows a sector AOB of a circle with centre O and radius r

4



The angle AOB is θ radians

The sector has an area of 9 cm^2 and perimeter 15 cm .

Find the value of θ and r

$$A = \frac{1}{2} r^2 \theta \Rightarrow \frac{1}{2} r^2 \theta = 9 \quad (1)$$

$$p = 2r + r\theta \Rightarrow 2r + r\theta = 15$$

$$\therefore r(2 + \theta) = 15 \quad (2)$$

$$\text{From (1)} \quad \theta = \frac{18}{r^2}$$

$$\therefore r \left(2 + \frac{18}{r^2} \right) = 15$$

$$2r + \frac{18}{r} = 15$$

$$2r^2 + 18 = 15r$$

$$2r^2 - 15r + 18 = 0$$

$$(2r - 3)(r - 6) = 0$$

$$\therefore r = \frac{3}{2} \quad \text{or} \quad r = 6$$

$$\therefore \theta = 8 \quad \text{or} \quad \theta = \frac{1}{2}$$

as $8 > 2\pi$ reject

$$\therefore r = 6 \text{ cm}, \quad \theta = \frac{1}{2}$$

Question 19

Adrian picks two notes at random from five \$5, three \$10 and two \$20 notes.

Find the probability that Adrian has picked a total of at least \$20.

3

Possible outcomes: \$5 + \$20, \$10 + \$10, \$10 + \$20, \$20 + \$20

$$P(5, 20) = \left(\frac{5}{10} \times \frac{2}{9} \right) \times 2 = \frac{20}{90}$$

$$P(10, 10) = \frac{3}{10} \times \frac{2}{9} = \frac{6}{90}$$

$$P(10, 20) = \left(\frac{3}{10} \times \frac{2}{9} \right) \times 2 = \frac{12}{90}$$

$$P(20, 20) = \frac{2}{10} \times \frac{1}{9} = \frac{2}{90}$$

$$\therefore P = \frac{20 + 6 + 12 + 2}{90}$$

$$= \frac{4}{9}$$

Question 20

Evaluate $\int_0^1 \frac{4x+2}{x^2+x+1} dx$.

2

$$I = \int_0^1 \frac{2(2x+1)}{x^2+x+1} dx$$

$$= 2 \left[\ln |x^2+x+1| \right]_0^1$$

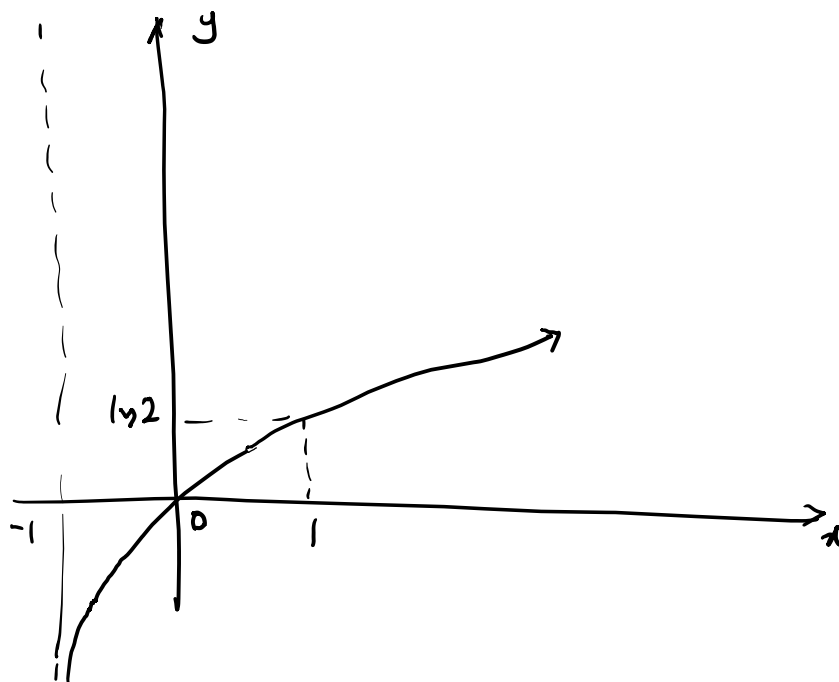
$$= 2 [\ln 3 - \ln 1]$$

$$= 2 \ln 3$$

Question 21

(a) Sketch the curve $y = \ln(x + 1)$, showing all important features.

2



(b) Using the trapezoidal rule with 4 subintervals, estimate the value of $\int_1^3 \ln(x + 1) dx$ correct to 3 decimal places

2

x	1	1.5	2	2.5	3
y	$\ln 2$	$\ln 2.5$	$\ln 3$	$\ln 3.5$	$\ln 4$

$$\therefore \int_1^3 \ln(x+1) dx \approx \frac{0.5}{2} [\ln 2 + \ln 4 + 2(\ln 2.5 + \ln 3 + \ln 3.5)]$$

$$\approx 2.154 \text{ (to 3 d.p.)}$$

Question 22

The amount of money gained by an entrepreneur doubles every day. On day 1, she earns \$9 and on day 2 she earns \$18.

3

On which day will her cumulative earnings first exceed \$1,000,000?

$$T_n = 9(2^{n-1})$$

$$\therefore \text{Cumulative earning: } S_n = \frac{9(2^n - 1)}{2 - 1}$$

$$= 9(2^n - 1)$$

$$\therefore 9(2^n - 1) > 1,000,000$$

$$2^n > \frac{1,000,000}{9} + 1$$

$$n > \log_2(111112.111\ldots)$$

$$\approx 16.76$$

By day 17

Question 23

Prove that $\frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} = 2 \operatorname{cosec} x$

3

$$\text{LHS} = \frac{\sin^2 x + (1 + \cos x)^2}{\sin x(1 + \cos x)}$$

$$= \frac{\sin^2 x + 1 + 2 \cos x + \cos^2 x}{\sin x(1 + \cos x)}$$

$$= \frac{2 + 2 \cos x}{\sin x(1 + \cos x)}$$

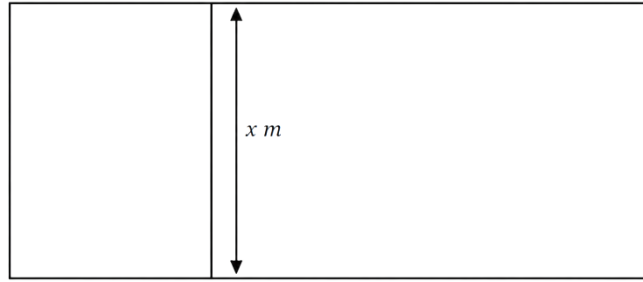
$$= \frac{2(1 + \cos x)}{\sin x(1 + \cos x)}$$

$$= \frac{2}{\sin x}$$

$$= \text{RHS}$$

Question 24

A farmer has a large patch of land, on which he needs to build two fenced-in, rectangular areas as shown in the diagram.



- (a) Given the farmer has 260m of fencing, show that the total area of the two rectangles is given by

1

$$A = 130x - \frac{3x^2}{2}$$

Let the horizontal length be L

$$\therefore 2L + 3x = 260$$

$$L = 130 - \frac{3x}{2}$$

Now $A = xL$

$$= x \left(130 - \frac{3x}{2} \right)$$

$$= 130x - \frac{3x^2}{2}$$

- (b) Hence or otherwise, find the maximum area of the paddock, showing that it is a maximum area.

3

$$A' = 130 - 3x$$

Stationary when $A' = 0$

$$\therefore 3x = 130$$

$$x = \frac{130}{3}$$

testing for maximum:

$$A'' = -3 < 0$$

$\therefore$ always concave down

$\therefore x = \frac{130}{3}$ gives maximum area

$$\therefore \text{max area} = 130 \left(\frac{130}{3} \right) - \frac{3 \left(\frac{130}{3} \right)^2}{2}$$

$$= \frac{8450}{3} \text{ m}^2$$

$$\approx 2816.7 \text{ m}^2$$

Question 25

In a survey of 200 students.

- 120 play soccer
- 90 play basketball
- 170 play at least one of the two sports.

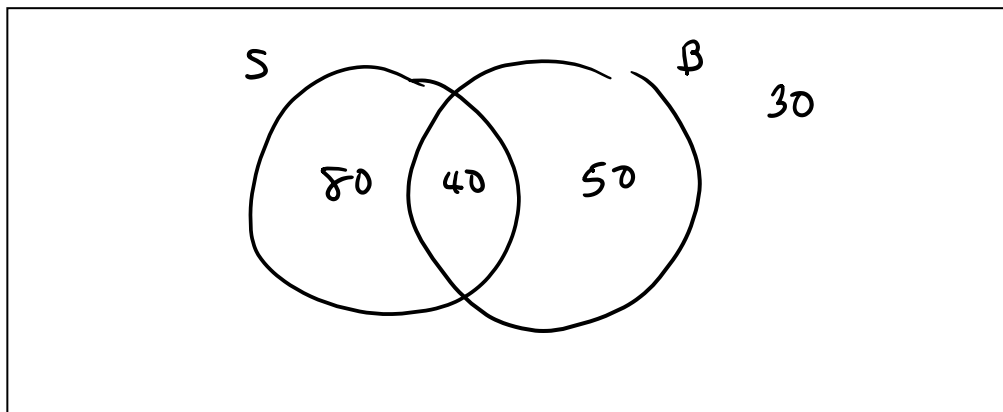
- (a) By finding the number of students that play both sports. Draw a Venn diagram showing all the relevant information.

2

$$n(S \cup B) = n(S) + n(B) - n(S \cap B)$$

$$\therefore 170 = 120 + 90 - n(S \cap B)$$

$$\therefore n(S \cap B) = 40$$



- (b) Two students are randomly selected without replacement. Find the probability that at least one of them plays exactly one of the two sports.

2

There are 130 students that does exactly one sport.

$$\therefore n(\text{not exactly one sport}) = 200 - 130 = 70$$

Probability at least two selected play exactly one sport:

$$1 - p(\text{neither selected plays exactly one sport})$$

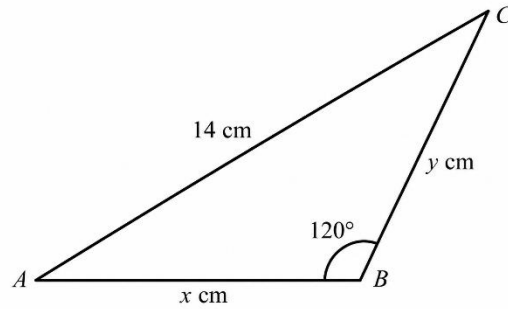
$$= 1 - \frac{70}{200} \cdot \frac{69}{199}$$

$$= 1 - \frac{4830}{39800}$$

$$= \frac{3497}{3980}$$

Question 26

In the triangle ABC below, $AB = x$ cm, $BC = y$ cm. $AC = 14$ cm and angle $ABC = 120^\circ$



(a) Show that $x^2 + y^2 + xy = 196$

1

Using the cosine rule

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos\angle ABC$$

$$14^2 = x^2 + y^2 - 2xy\cos 120^\circ$$

$$\text{Since } \cos 120^\circ = -\frac{1}{2}$$

$$196 = x^2 + y^2 + xy$$

$$\text{i.e. } x^2 + y^2 + xy = 196$$

(b) Given that the perimeter of triangle ABC is 30 cm, find the exact area of triangle ABC

3

$$\therefore \text{Perimeter} = 30 \text{ cm} \Rightarrow x + y + 14 = 30$$

$$\therefore x + y = 16$$

$$\therefore x^2 + 2xy + y^2 = 256$$

subtract expression from part (a)

$$xy = 60$$

$$\therefore A = \frac{1}{2} xy \sin 120^\circ$$

$$= \frac{1}{2} \times 60 \times \frac{\sqrt{3}}{2}$$

$$= 15\sqrt{3} \text{ cm}^2$$

Question 27

Let $f(x) = \frac{1}{e^x + 1}$ and $g(x) = f(x) - \frac{1}{2}$

3

Prove that $g(x)$ is an **odd** function

$$g(-x) = f(-x) - \frac{1}{2}$$

$$= \frac{1}{e^{-x} + 1} - \frac{1}{2}$$

$$= \frac{e^x}{1 + e^x} - \frac{1}{2}$$

$$= \frac{2e^x - (1 + e^x)}{2(1 + e^x)}$$

$$= \frac{e^x - 1}{2(1 + e^x)}$$

Now $-g(x) = \frac{1}{2} - f(x)$

$$= \frac{1}{2} - \frac{1}{e^x + 1}$$

$$= \frac{e^x + 1 - 2}{2(e^x + 1)}$$

$$= \frac{e^x - 1}{2(e^x + 1)}$$

$$= g(-x)$$

Question 28

Solve $\ln(6e^x + 1) + x = 0$, writing your answer in logarithmic form

3

$$\ln(6e^x + 1) = -x$$

$$\therefore 6e^x + 1 = e^{-x}$$

$$6e^{2x} + e^x = 1$$

$$6e^{2x} + e^x - 1 = 0$$

$$\text{let } u = e^x$$

$$\therefore 6u^2 + u - 1 = 0$$

$$(3u - 1)(2u + 1) = 0$$

$$\therefore u = \frac{1}{3} \text{ or } u = -\frac{1}{2}$$

$$e^x = \frac{1}{3} \quad \text{or} \quad e^x = -\frac{1}{2}$$

$$x = \ln\left(\frac{1}{3}\right) \quad \therefore x \text{ has no solutions}$$

$$= -\ln 3$$

Question 29

A particle, P , travels in a straight line from O such that its velocity, v m/s, at time t seconds after passing O is given by $v = 24 \left(e^{-\frac{t}{6}} - \frac{1}{e} \right)$

- (a) Find the value of t at which the particle P is instantaneously at rest.

2

$$\text{at rest: } v=0 \Rightarrow 24 \left(e^{-\frac{t}{6}} - \frac{1}{e} \right) = 0$$

$$e^{-\frac{t}{6}} - \frac{1}{e} = 0$$

$$e^{-\frac{t}{6}} = \frac{1}{e}$$

$$-\frac{t}{6} = \ln \left(\frac{1}{e} \right)$$

$$= -1$$

$$t = 6$$

- (b) Find the total distance travelled by the particle P for the first 9 seconds to the nearest metre.

3

$$x = 24 \int e^{-\frac{t}{6}} - \frac{1}{e} dt$$

$$= 24 \left(-6 \cdot e^{-\frac{t}{6}} - \frac{1}{e} t \right) + C$$

$$= -144 e^{-\frac{t}{6}} - \frac{24}{e} t + C$$

$$\text{when } t=0 \quad x = -144 + C$$

$$\text{when } t=6 \quad x = -144 e^{-1} - \frac{144}{e} + C$$

$$= -\frac{288}{e} + C$$

$$\text{when } t=9 \quad x = -144 e^{-\frac{3}{2}} - \frac{216}{e} + C$$

$$\therefore \text{distance travelled from } t=0 \text{ to } t=6 : 144 - \frac{288}{e}$$

$$\text{distance from } t=6 \text{ to } 9 : 144 e^{-\frac{3}{2}} - \frac{72}{e}$$

$$\therefore \text{total distance: } 144 - \frac{360}{e} + 144 e^{-\frac{3}{2}}$$

$$\approx 44 \text{ m (to nearest metre)}$$

Question 30

(a) Show that $\frac{d}{dx} \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right) = \frac{2}{(\sin x + \cos x)^2}$

2

$$u = \sin x - \cos x$$

$$v = \sin x + \cos x$$

$$u' = \cos x + \sin x$$

$$v' = \cos x - \sin x$$

$$\begin{aligned} \therefore \frac{d}{dx} \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right) &= \frac{(\sin x + \cos x)^2 + (\sin x - \cos x)^2}{(\sin x + \cos x)^2} \\ &= \frac{\sin^2 x + 2\sin x \cos x + \cos^2 x + \sin^2 x - 2\sin x \cos x + \cos^2 x}{(\sin x + \cos x)^2} \\ &= \frac{2(\sin^2 x + \cos^2 x)}{(\sin x + \cos x)^2} \\ &= \frac{2}{(\sin x + \cos x)^2} \end{aligned}$$

(b) Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin x - \cos x + 2}{(\sin x + \cos x)^2} dx$

3

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \frac{\sin x - \cos x}{(\sin x + \cos x)^2} + \frac{2}{(\sin x + \cos x)^2} dx \\ &= - \int_0^{\frac{\pi}{4}} \frac{\cos x - \sin x}{(\sin x + \cos x)^2} dx + \int_0^{\frac{\pi}{4}} \frac{2}{(\sin x + \cos x)^2} dx \\ &= \left[\frac{1}{\sin x + \cos x} \right]_0^{\frac{\pi}{4}} + \left[\frac{\sin x - \cos x}{\sin x + \cos x} \right]_0^{\frac{\pi}{4}} \\ &= \left[\left(\frac{1}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}} \right) - \left(\frac{1}{1} \right) \right] + \left[\left(\frac{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}}{0 + 1} \right) - \left(\frac{-1}{1} \right) \right] \\ &= \left[\frac{\sqrt{2}}{2} - 1 \right] + (1) \\ &= \frac{\sqrt{2}}{2} \end{aligned}$$

Question 31

Given the function $f(x) = \ln(x^2 + 1) - x$

- (a) Find the stationary point(s) and determine their nature.

3

$$f'(x) = \frac{2x}{x^2+1} - 1$$

stationary point when $f'(x) = 0$

$$\therefore \frac{2x}{x^2+1} - 1 = 0$$

$$\therefore \frac{2x}{x^2+1} = 1$$

$$\therefore x^2 + 1 = 2x$$

$$x^2 - 2x + 1 = 0$$

$$\therefore (x-1)^2 = 0$$

$$x = 1$$

$$\text{when } x = 1 \quad y = \ln 2 - 1$$

Testing the nature

x	0	1	2
y'	-1	0	$-\frac{1}{5}$

1 - 1

$\therefore (1, \ln 2 - 1)$ is a horizontal point of inflection

(b) Find the point(s) of inflection.

2

$$\begin{aligned}
 f''(x) &= \frac{(x^2+1) \cdot 2 - 2x \cdot 2x}{(x^2+1)^2} \\
 &= \frac{2x^2+2-4x^2}{(x^2+1)^2} \\
 &= \frac{2-2x^2}{(x^2+1)^2} \\
 &= \frac{2(1-x)(1+x)}{(x^2+1)^2}
 \end{aligned}$$

$\therefore$ possible point of inflection when $f''(x)=0$

$$\Rightarrow x=1 \quad \text{or} \quad x=-1$$

$$\therefore y = \ln 2 - 1 \quad \text{or} \quad y = \ln 2 + 1$$

From (a) $(1, \ln 2 - 1)$ is a horizontal point of inflection

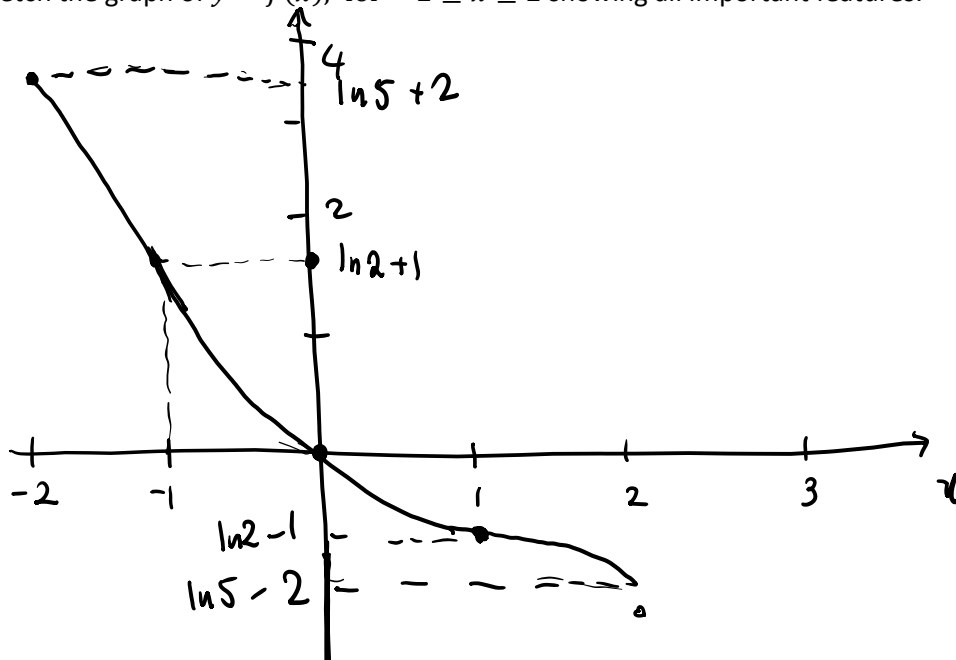
Testing $x = -1$

x	-2	-1	0
$f''(x)$	$-\frac{6}{25}$	0	2
	$\cap$	.	$\cup$

$\therefore (-1, \ln 2 + 1)$ is a point of inflection

(c) Sketch the graph of $y = f(x)$, for $-2 \leq x \leq 2$ showing all important features.

2



Question 32

4

The temperature T °C inside a greenhouse is modelled by

$$T = 20 - 8 \cos\left(\frac{\pi}{12}(t - 4)\right)$$

where t is the number of hours after midnight.

An automatic shade cloth is deployed whenever the temperature exceeds 26°C.

Determine the total amount of time each day that the shade cloth is deployed.

$$T = \frac{2\pi}{n} = \frac{2\pi}{\frac{\pi}{12}}$$

$$= 24 \text{ hours}$$

$$\therefore \text{so domain: } 0 \leq t \leq 24$$

$$-\frac{\pi}{3} \leq \frac{\pi}{12}(t-4) \leq \frac{5\pi}{3}$$

Solve for equality:

$$20 - 8 \cos\left(\frac{\pi}{12}(t-4)\right) = 26$$

$$\cos\left(\frac{\pi}{12}(t-4)\right) = -\frac{3}{4}$$

$\therefore$ in quadrants 2 and 3

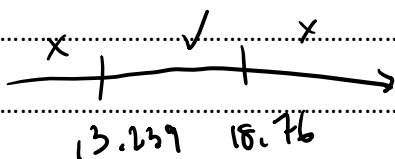
$$\therefore \frac{\pi}{12}(t-4) = \pi - \cos^{-1}\left(\frac{3}{4}\right) \text{ or } \pi + \cos^{-1}\left(\frac{3}{4}\right)$$

$$t-4 = 12 - \frac{12 \cos^{-1}\left(\frac{3}{4}\right)}{\pi} \text{ or } 12 + \frac{12 \cos^{-1}\left(\frac{3}{4}\right)}{\pi}$$

$$t = 16 - \frac{12 \cos^{-1}\left(\frac{3}{4}\right)}{\pi} \text{ or } 16 + \frac{12 \cos^{-1}\left(\frac{3}{4}\right)}{\pi}$$

$$\approx 13.239 \text{ or } 18.761$$

Testing region:



$$\text{Test } t = 14$$

$T = 26.9$ which satisfies the inequality

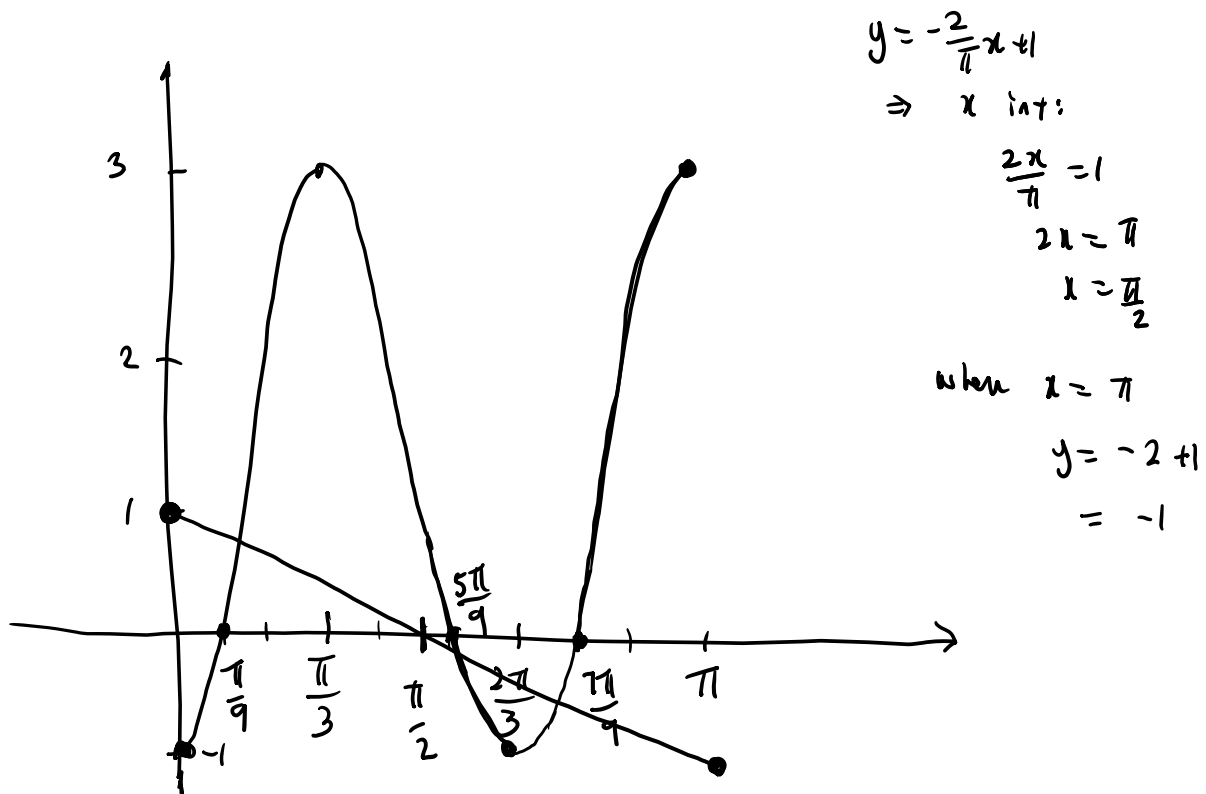
$$\therefore \text{total time} \approx 18.761 - 13.239$$

$$\approx 5.52 \text{ hours}$$

Question 33

3

- (a) Sketch the graphs of $y = -2 \cos 3x + 1$ and $y = -\frac{2}{\pi}x + 1$ on the same axes, for $0 \leq x \leq \pi$



$$\frac{5\pi}{9} \approx 1.75$$

$$\frac{7\pi}{9} \approx 1.57$$

- (b) Explain how the number of solutions for $2 \cos 3x = \frac{2}{\pi}x$ can be found using the graph drawn in part (a)

1

The solutions to $2 \cos 3x = \frac{2}{\pi}x$ are equivalent to the solutions to $2 \cos 3x + 1 = \frac{2}{\pi}x + 1$

$\therefore$ we can see the number of solutions by looking at how many intersection points on the curve drawn in part (a)

Question 34

4

Let g be the function obtained from f by a sequence of transformations such that

$$g(x) = af(b-x) + c.$$

The point $(2, 5)$ on the graph of $y = f(x)$ corresponds to the point $(7, -11)$ on the graph of $y = g(x)$.

The point $(4, 1)$ on the graph of $y = f(x)$ corresponds to the point $(5, -3)$ on the graph of $y = g(x)$.

Find the values of a , b and c .

$$\frac{y-c}{a} = f(-x+b)$$

$g(x)$ is transformed in the following sequence:

- vertical dilation by factor of a
- shifted up by c units
- shifted left by b units
- flipped around y axis.

Applying transformation on $(2, 5)$ yields

$$[-(2-b), 5a+c]$$

$$= [b-2, 5a+c] \quad \therefore b-2=7$$

$$\therefore \boxed{b=9}$$

$$\text{and } 5a+c=-11$$

(1)

Applying transformation on $(4, 1)$ yields

$$(-(4-b), a+c)$$

$$b-4=5 \quad \text{and} \quad a+c=-3$$

(2)

$$b=9$$

$$(1) - (2)$$

$$4a = -8$$

$$a = -2$$

$$\therefore c = -1$$

$$\therefore \boxed{a = -2, b = 9 \text{ and } c = -1}$$

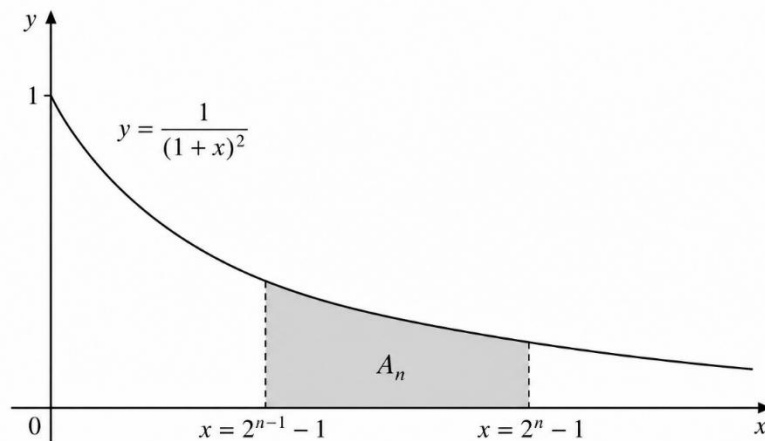
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Question 35

Let $f(x) = \frac{1}{(1+x)^2}, x \geq 0$.

The shaded region A_n is bounded by the curve $y = f(x)$, the x -axis, and the vertical lines

$x = 2^{n-1} - 1$ and $x = 2^n - 1$ as shown below.



(a) Show that $A_n = \frac{1}{2^n}$

2

$$A_n = \int_{2^{n-1}-1}^{2^n-1} (1+x)^{-2} dx$$

$$= \left[- (1+x)^{-1} \right]_{2^{n-1}-1}^{2^n-1}$$

$$= - \left[\frac{1}{1+x} \right]_{2^{n-1}-1}^{2^n-1}$$

$$= - \left[\frac{1}{1+2^n-1} - \frac{1}{1+2^{n-1}-1} \right]$$

$$= - \left(\frac{1}{2^n} - \frac{1}{2^{n-1}} \right)$$

$$= \frac{2}{2^n} - \frac{1}{2^n}$$

$$= \frac{1}{2^n}$$

(b) Show that $A_n = A_{n+1} + A_{n+2} + A_{n+3} + \dots$

2

$$A_{n+1} = \frac{1}{2^{n+1}}, A_{n+2} = \frac{1}{2^{n+2}}, A_{n+3} = \frac{1}{2^{n+3}}$$

$$\therefore \text{RHS} = \frac{1}{2^{n+1}} + \frac{1}{2^{n+2}} + \frac{1}{2^{n+3}} + \dots$$

this is a limiting sum with $a = \frac{1}{2^{n+1}}$ and $r = \frac{1}{2}$

$$\therefore = \frac{\frac{1}{2^{n+1}}}{1 - \frac{1}{2}}$$

$$= \frac{1}{2^{n+1}} \times \frac{1}{\frac{1}{2}}$$

$$= \frac{1}{2^n}$$

$$= A_n \text{ (from part a)}$$

(c) Find the value of S given that $S = A_1 + 2A_2 + 3A_3 + 4A_4 + \dots$

3

$$A_1 = \frac{1}{2^1}, A_2 = \frac{1}{2^2}, A_3 = \frac{1}{2^3}, \dots$$

$$\therefore S = A_1 + 2A_2 + 3A_3 + 4A_4 + \dots$$

$$= (A_1 + A_2 + A_3 + A_4 + \dots) + (A_2 + A_3 + A_4 + \dots) + (A_3 + A_4 + A_5 + \dots) + \dots$$

$$= \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots \right) + \left(\frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots \right) + \left(\frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \dots \right) + \dots$$

$$\text{A.P. } a = \frac{1}{2}, r = \frac{1}{2}$$

$$\text{A.P. } a = \frac{1}{2^2}, r = \frac{1}{2} \text{ etc.}$$

$$= \frac{\frac{1}{2}}{1 - \frac{1}{2}} + \frac{\frac{1}{4}}{1 - \frac{1}{2}} + \frac{\frac{1}{8}}{1 - \frac{1}{2}} + \dots$$

$$= 2 \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right)$$

$$= 2 \left(\frac{\frac{1}{2}}{1 - \frac{1}{2}} \right)$$

$$= 2$$